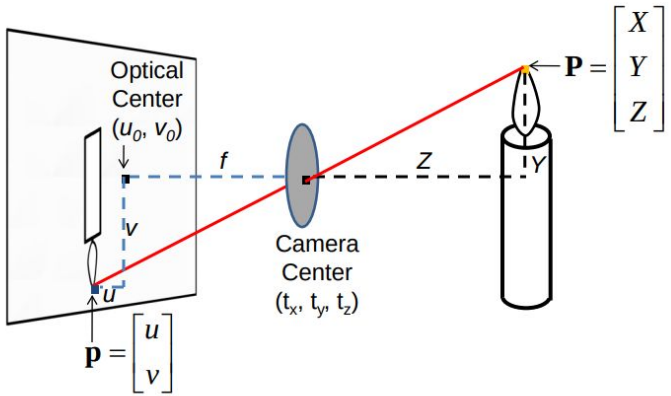
Pairwise Patch Extraction

Vishwesh

Code: https://github.com/Vishwesh4/Ink-WSI/tree/master/modules/register

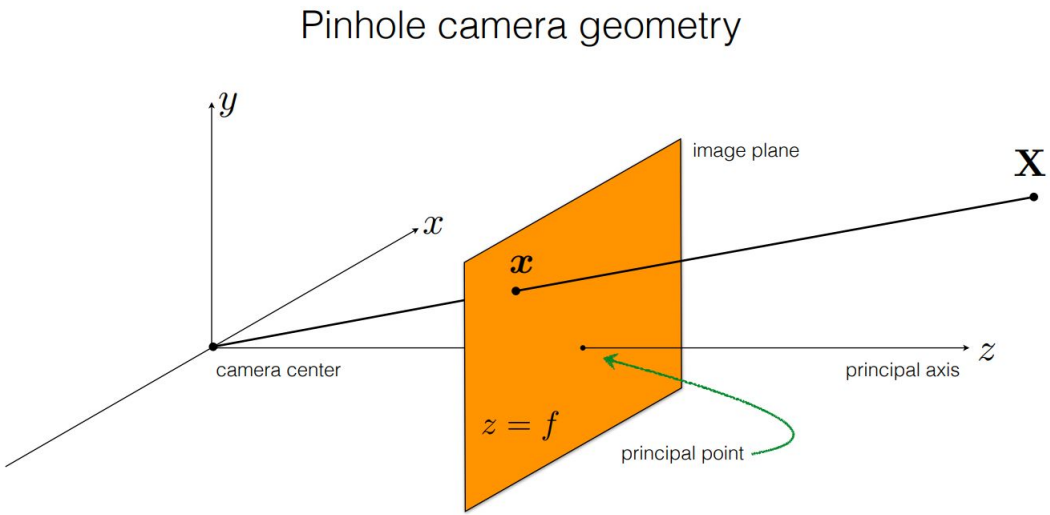
Background

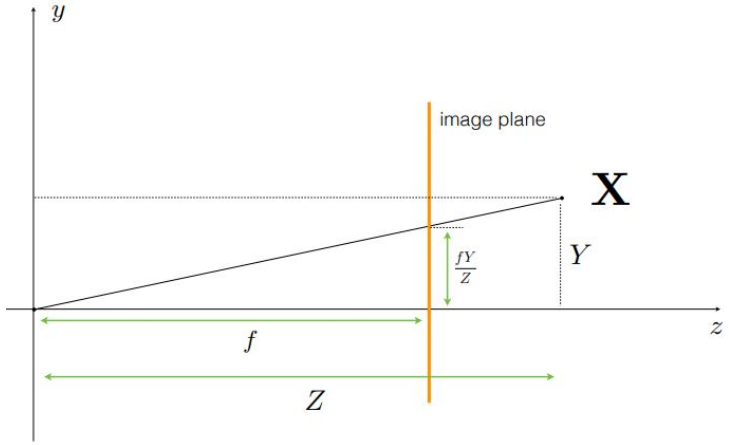
Projection: world coordinates→image coordinates



https://courses.engr.illinois.edu/cs543/sp2011/lectures/Lecture%2002%20-%20Projective%20Geometry%20and%20Camera%20Models%20-%20Vision_Spring2011.pdf

Background





$$[X \quad Y \quad Z]^\top \mapsto [fX/Z \quad fY/Z]^\top$$

What is the camera matrix **P** *for a pinhole camera model?*

https://www.cs.cmu.edu/~16385/s17/Slides/11.1_Camera_matrix.pdf



But we dont like non linear equations

Non linear to linear

Homogeneous coordinates

Conversion

Converting to *homogeneous* coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad (x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

homogeneous image coordinates

homogeneous scene coordinates

Converting *from* homogeneous coordinates

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w) \qquad \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Think of w as scale, smaller w implies closer distance hence x,y will blow up

Background

A camera is a mapping between the 3D world and a 2D image

$$x = PX$$

2D image point

camera matrix

3D world point

$$P = K[R|t]$$

https://www.cs.cmu.edu/~16385/s17/Slides/11.1_Camera_matrix.pdf

https://www.cs.cmu.edu/~16385/s17/Slides/11.1_Camera_matrix.pdf

Background

$$\boldsymbol{x} = \mathbf{P}\mathbf{X}$$

We added an extra dimension so as to linearize the equations, other wise it would be non linear

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

homogeneous image 3 x 1

Camera matrix 3 x 4

homogeneous world point 4 x 1

What is registration?

$$x = PX$$

2D image point | camera matrix | 3D world point

$$x = PX$$

2D image point | camera matrix | 3D world point

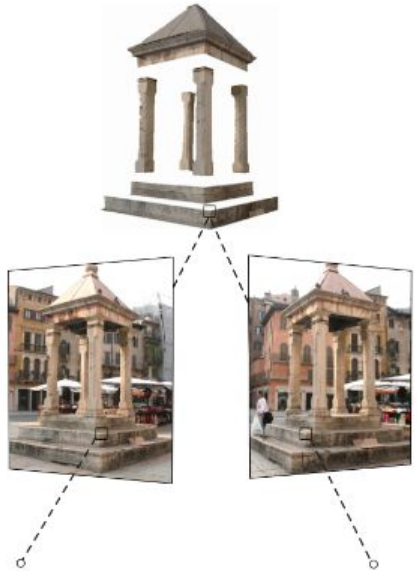


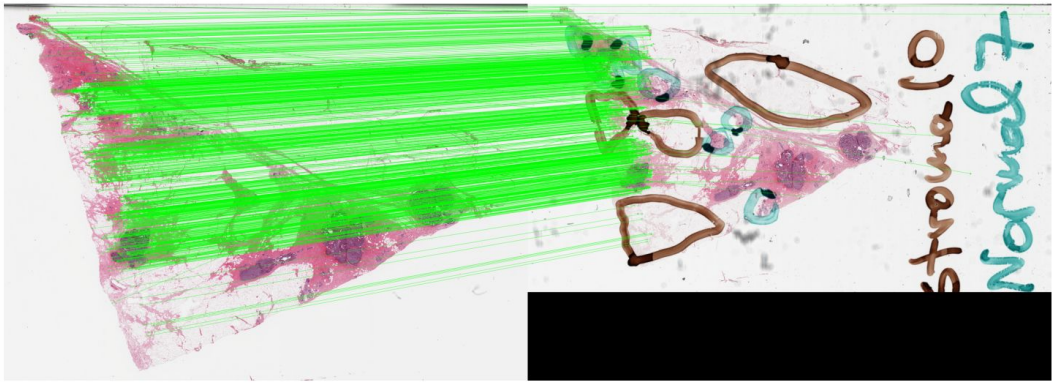
Figure 4. Two perspective views of the same 3-D scene with two conjugate points singled out.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

How is it usually done?

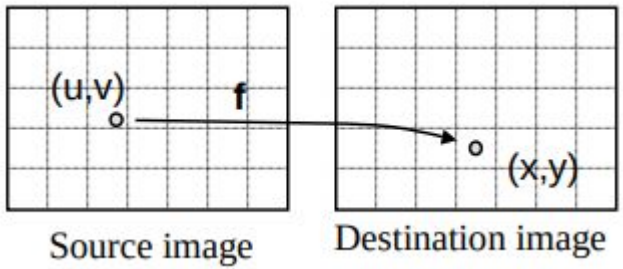
We have x’ and x from source and target images, we need 9 pairs of x’ and x to solve.

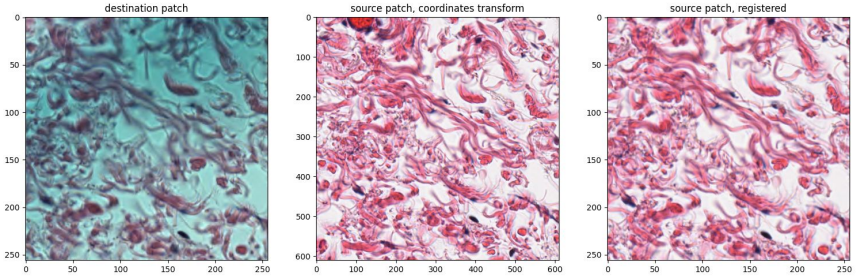
We use SIFT to get pairs, and use that to solve for H given 9 pairs. Since SIFT can have lot of errors, RANSAC is used for getting good set of 9 pairs

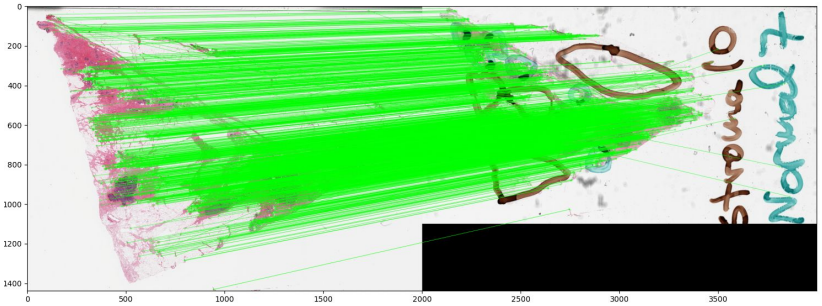


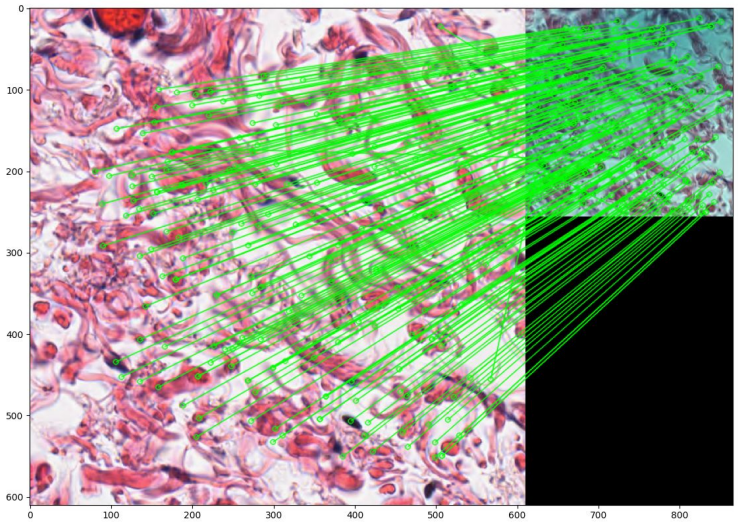
Pseudo Code

1. Downsample target and source image to required sizes (if needed)
2. Use SIFT to get pair of points
3. Use RANSAC to solve for transformation matrix
4. Perform image warping to transform source image to target image




destination patch
source patch, coordinates transform
source patch, registered





What to do when image is too big for image warping

- Image warping works by sampling points from matrix and applying transformation matrix on each of the points
- Whole slide images are too big for this!!!!!!

We need a method to extract patch from source whole slide image and register it to target patch from target whole slide image, without needed to do SIFT and RANSAC again and again

transformation successfully. Let $\mathbf{M} \in \mathbb{R}^{3\times3}$ be the transformation matrix, $\mathbf{x}^s \in \mathbb{R}^{2\times1}$ and $\mathbf{x}^t \in \mathbb{R}^{2\times1}$, be two points in source and target slides, we have the projective transformation between points defined as,

$$\mathbf{x}_1^t = \frac{\mathbf{M}_{1:}[\mathbf{x}^s\,;\,1]}{\mathbf{M}_{3:}[\mathbf{x}^s\,;\,1]} \qquad (2)$$
$$\mathbf{x}_2^t = \frac{\mathbf{M}_{2:}[\mathbf{x}^s\,;\,1]}{\mathbf{M}_{3:}[\mathbf{x}^s\,;\,1]}$$

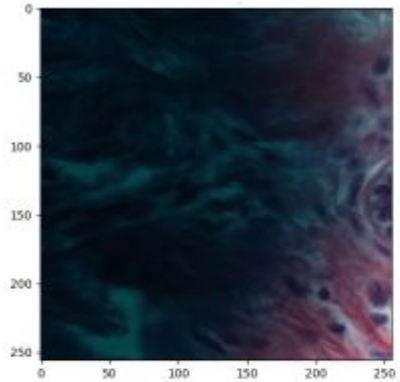
Let $\mathbf{s} \in \mathbb{R}^{2\times1}$ and $\mathbf{t} \in \mathbb{R}^{2\times1}$, be two corresponding points using which patches were extracted from source and target slides respectively. We wish to find the new transformation matrix for the source and target patches, extracted using these points as reference. Hence, we wish to find $\mathbf{M}' \in \mathbb{R}^{3\times3}$ such that

$$\mathbf{x}_1^t - \mathbf{t}_1 = \frac{\mathbf{M}'_{1:}[\mathbf{x}^s - \mathbf{s}\,;\,1]}{\mathbf{M}'_{3:}[\mathbf{x}^s - \mathbf{s}\,;\,1]} \qquad (3)$$
$$\mathbf{x}_2^t - \mathbf{t}_2 = \frac{\mathbf{M}'_{2:}[\mathbf{x}^s - \mathbf{s}\,;\,1]}{\mathbf{M}'_{3:}[\mathbf{x}^s - \mathbf{s}\,;\,1]}$$

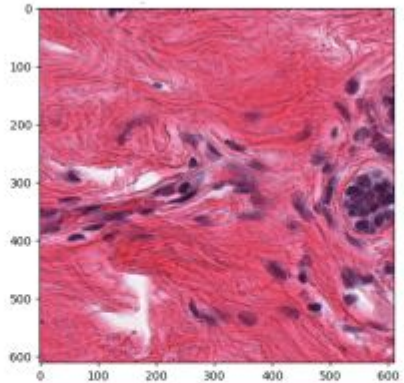
$$M' = \begin{bmatrix} M_{11} - M_{31}t_1 & M_{12} - M_{32}t_1 & M_{13} - M_{33}t_1 + s_1(M_{11} - M_{31}t_1) + s_2(M_{12} - M_{32}t_1) \\ M_{21} - M_{31}t_2 & M_{22} - M_{32}t_2 & M_{23} - M_{33}t_2 + s_1(M_{21} - M_{31}t_2) + s_2(M_{22} - M_{32}t_2) \\ M_{31} & M_{32} & M_{33} + M_{31}s_1 + M_{32}s_2 \end{bmatrix}$$

This way, we just need to do the following

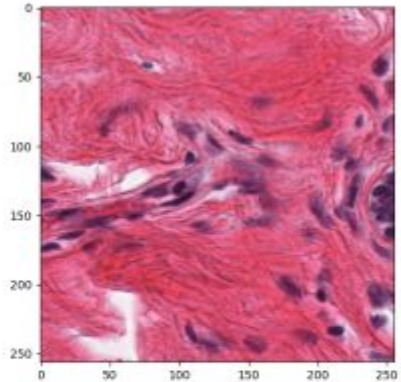
1. Find M from sift and ransac using the whole slide
2. Extract patch at point of interest from source slide. Let that point be s
3. Find t using M, and extract patch at corresponding point t from target slide
4. Find new transformation matrix M' for the extracted pair of patch
5. Warp the source image patch using M'
6. You are done!!!!!! This is very fast and accurate too!! compared to doing registration for each patch



(a) Extracted patch from target slide (ink marked slide) at point $\mathbf{t}$



(b) Extracted patch from source slide (clean slide) at point $\mathbf{s}$



(c) Projective transformation using $\mathbf{M}'$ from fig. 4b to fig. 4a